Supplementary Material

A. Broader Impacts

Our research on TIE aims to enhance semantic fidelity in complex, multi-concept prompts, enabling more accurate alignment between text and image. This improvement supports a wide range of applications, including educational visualizations, assistive design tools, and creative authoring for artists and storytellers. At the same time, by making generative models more reliable, we also lower the barrier to malicious uses such as highly realistic deepfakes for disinformation, fraud, and privacy or copyright infringement. Even when functioning as intended, subtle misbindings (e.g., swapping or omitting attributes) in critical contexts such as medical illustrations or legal diagrams could mislead users, and erroneous outputs may go unnoticed in black-box deployments.

B. Implementation details

Analysis and Method Details. We conduct all experiments on a single NVIDIA RTX A6000 GPU. Our TIE builds upon SDXL [12] and SD3.5 [3]. SDXL relies on CLIP text embeddings, which enables clear analysis of the effects of CLIP text embedding manipulation. Therefore, we use SDXL as the primary backbone in our analysis. SD3.5 employs multiple text encoders, including OpenCLIP ViT-G, CLIP ViT-L, and T5-XXL. In this setting, we apply our method only to the CLIP text embeddings. We also employ the Stanza package for automatic parsing.

Hyperparameters. The scaling factor is fixed to $\delta = 1.2$ for all backbones. The hyperparameter C controls the amount of the mean embedding of preceding concepts subtracted from each subsequent concept embedding. For SDXL, we use $(\gamma, \lambda_{\text{ext}}, \lambda_{\text{int}}, C) = (0.0005, 0.1, 0.2, 0.5)$, which provides the best overall balance across the five evaluation categories. For SD3.5-Medium, we use 28 inference steps with a guidance scale of 4.5 and $(\gamma, \lambda_{\text{ext}}, \lambda_{\text{int}}, \mathcal{C}) = (0.001, 0.1, 0.1, 0.9)$. For SD3.5-Large, we use 28 inference steps with a guidance scale of 3.5 and $(\gamma, \lambda_{\text{ext}}, \lambda_{\text{int}}, \mathcal{C}) = (0.0005, 0.1, 0.2, 0.5)$.

Concept Parsing and Multi-token Aggregation. We automatically extract attribute–object pairs using the Stanza dependency parser and align each extracted concept with its corresponding CLIP token span. Let $\mathcal{T}(q)$ denote the CLIP token indices associated with concept q. For a prompt with K parser-matched pairs, we define related pairs as $\mathcal{R} = \{(a_k, o_k)\}_{k=1}^{K}$ and unrelated pairs as $\mathcal{U} = \{(a_k, o_l) : k \neq l\}$. When a concept is split into multiple CLIP tokens, we treat the complete span as a single concept and average its token-level quantities. In particular, its concept-level entropy is

$$H(q) = \frac{1}{|\mathcal{T}(q)|} \sum_{i \in \mathcal{T}(q)} H_i,$$

where H_i is defined in Eq. (1) of the main paper.

Attention Aggregation. Let $A_{i,j}^{(h)}$ denote the Layer-0 attention from query token i to key token j at head h. We first average over attention heads, $\bar{A}_{i,j} = \frac{1}{N_h} \sum_{h=1}^{N_h} A_{i,j}^{(h)}$. Since CLIP uses causal self-attention, we aggregate only causally valid cross-concept entries. For concepts q and r, let

$$\begin{aligned}\mathcal{V}(q,r) = &\{(i,j) : i \in \mathcal{T}(q), j \in \mathcal{T}(r), j \leq i\} \\ &\cup \{(i,j) : i \in \mathcal{T}(r), j \in \mathcal{T}(q), j \leq i\}.\end{aligned}$$

The concept-level attention score is then

$$A(q,r) = \frac{1}{|\mathcal{V}(q,r)|} \sum_{(i,j) \in \mathcal{V}(q,r)} \bar{A}_{i,j}.$$

This provides an order-independent score while respecting CLIP's causal mask and is used as the binding strength in IEB.

Multiple Concept Pairs. For prompts containing multiple concepts, AEP is first applied to all extracted concepts using their aggregated entropy. IEB then interpolates every related pair $(a_k, o_k) \in \mathcal{R}$ using $\alpha_k = \lambda_{\text{int}}(1 - A(a_k, o_k))$, and extrapolates every unrelated pair $(a_k, o_l) \in \mathcal{U}$ using $\beta_{k,l} = \lambda_{\text{ext}} A(a_k, o_l)$. Thus, the two-pair formulation in the main paper is used only for exposition; the implementation supports an arbitrary number of parser-resolved concept pairs. All operations are performed once before diffusion sampling.

C. Datasets

We use four types of datasets for our analysis and experiments.

Synthetic Dataset. We conduct our token-entropy analysis (Sec 3.1b) on synthetic prompts sampled from the A&E [2] dataset. The left plot used 64 prompts of the object category from the A&E dataset and switched position of the attribute and objects, to compare the entropy of the tokens when the prompt takes the form of "a ⟨attr$_1$⟩ ⟨obj$_1$⟩ and a ⟨attr$_2$⟩ ⟨obj$_2$⟩" with when it takes the form of "a ⟨attr$_2$⟩ ⟨obj$_2$⟩ and a ⟨attr$_1$⟩ ⟨obj$_1$⟩". For the right plot, we sampled attributes and objects from the same 64 prompts and made prompts in the form of "a ⟨attr$_1$⟩ ⟨obj$_1$⟩ and a ⟨attr$_2$⟩ ⟨obj$_2$⟩ and a ⟨attr$_3$⟩ ⟨obj$_3$⟩".

A&E. For Sec. 3.2, we use the A&E [2] dataset consisting of 64 prompts across various object categories. Each prompt follows the format: "a ⟨attr$_1$⟩ ⟨obj$_1$⟩ and a ⟨attr$_2$⟩ ⟨obj$_2$⟩". To further investigate with more diverse sentence structures and datasets, we additionally conduct experiments as shown in Sec. E.1.

T2I-Compbench. We evaluate the attribute types of Color, Shape, and Texture, Spatial and Complex using the T2I CompBench [8]. Specifically, Complex category consists of prompts containing more than two attributes and objects and involving at least two of the following subcategories: Color, Shape, Texture and object relationships. Following prior works [4–6, 9, 14], we randomly select 60 long prompts and 240 short prompts for each category, resulting in 300 prompts per attribute type. For each prompt, we generate images using five different random seeds for evaluation.

Concept Conjunction 500 (CC-500). Each prompt consists of two subjects, with one attribute assigned to each. Following the setup in [15], we categorize the prompts into three types: two living subjects, one living and one non-living subject, and two non-living subjects. We randomly select 80 prompts per category, resulting in 240 prompts, and generate 10 images for each.

GenEval2 [10] Evaluation Metrics Details. VQA [11] assesses prompt alignment by querying generated images with visual questions derived from the prompt. TIFA [7] strengthens this evaluation by decomposing each prompt into discrete atoms and aggregating VQA-based correctness across these atoms. SoftTIFA further extends TIFA by assigning continuous scores to each atom rather than binary judgments. In our evaluation, we report SoftTIFA$_{\text{AM}}$, the arithmetic mean over atom-level scores, and SoftTIFA$_{\text{GM}}$, the geometric mean that more strictly penalizes missing any required atom.

D. Hyperparameter Sensitivity

We evaluate the sensitivity of TIE to its four hyperparameters, γ, λ_{ext}, λ_{int}, and $\mathcal{C}$, on both SD3.5-Medium and SD3.5-Large. As shown in Tab. 1 and Tab. 2, TIE remains stable across a range of hyperparameter configurations, with only small variations in the average BLIP-VQA score. For SD3.5-Medium, the average score ranges from 0.6622 to 0.6667, while for SD3.5-Large it varies only from 0.6714 to 0.6729. Although individual categories may favor slightly different settings, the default configurations achieve the stable average performance on both models, providing a balanced improvement across the five evaluation categories. These results indicate that TIE does not rely on a narrowly tuned hyperparameter configuration and generalizes well to nearby parameter choices.

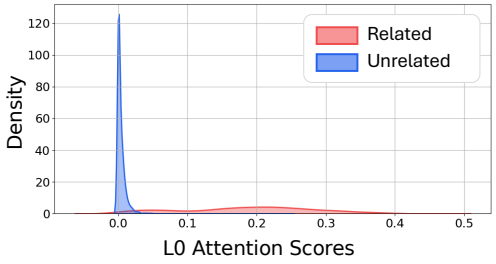


Figure 1. **L0 attention distinguishes related from unrelated token pairs.** The density of L0 attention scores is clearly shifted toward larger values for related attribute–object pairs, while unrelated pairs are concentrated near zero.

E. Analysis of the CLIP text encoder

E.1. L0 Attention Scores Analysis

Masking L0 Attention Scores. We find that L0 attention scores play a critical role in attribute binding across various T2I diffusion models in Sec. 3.2. To investigate the consistency of this behavior across various architectures, we extend the L0 attention score masking to an additional model. As shown in Fig. 6, we mask each key token corresponding to attributes in the first layer self-attention map of the SDXL model. This leads to the disappearance of the corresponding attributes in the generated images. Interestingly, the model frequently shows the removal of the attribute together with the associated object. This indicates that L0 attention scores are essential for establishing attribute binding as well as for preserving the presence of the related object, even in complex sentence structures. Furthermore, we find that these scores play a consistent role across various types of attributes, including color, shape, and texture.

L0 Attention Scores and Generated Images. We observe a correlation between L0 attention scores and the degree of semantic binding in the generated images, as shown in Fig. 3c. The L0 attention scores of related pairs are primarily distributed between 0 and 0.4, as illustrated in the main paper Fig.3c and supplementary Fig. 3. Based on this observation, we analyze the relationship between L0 attention scores and the generated images across different score intervals in Fig. 3. When the attention score $A_{i,j} < 0.1$, the corresponding concept often fails to match semantically, or the object does not appear. In the range $0.1 \leq A_{i,j} < 0.2$, the attribute may appear, but it is often entangled or ambiguously associated with the object. For higher scores, semantic binding between the attribute and object tends to improve substantially. These findings suggest that higher L0 attention scores generally indicate stronger concept bind-

Table 1. **Hyperparameter sensitivity of TIE based on SD3.5-Medium.** We report BLIP-VQA scores under different choices of γ, λ_{ext}, λ_{int}, and $\mathcal{C}$. The default configuration is highlighted in gray. Best results are shown in **bold** and second-best results are underlined.

Hyperparameters				BLIP-VQA ↑					
γ	λ_{ext}	λ_{int}	$\mathcal{C}$	Color	Shape	Texture	Spatial	Complex	Avg.
0.0005	0.1	0.2	0.5	0.7946	0.6079	0.7536	**0.6619**	0.4992	0.6634
0.0005	0.1	0.3	0.5	0.7930	0.6059	0.7523	**0.6619**	0.4978	0.6622
0.0005	0.1	0.1	0.5	0.7949	0.6079	0.7547	**0.6619**	0.4987	0.6636
0.0005	0.1	0.1	0.7	0.7967	0.6075	0.7560	0.6611	0.5000	0.6643
0.0005	0.1	0.1	0.9	**0.7978**	**0.6175**	0.7574	0.6604	0.5002	**0.6667**
0.0005	0.2	0.1	0.9	0.7976	0.6153	0.7568	0.6603	**0.5003**	0.6661
0.001	0.1	0.1	0.9	0.7938	0.6120	**0.7603**	0.6586	0.4986	0.6647

Table 2. **Hyperparameter sensitivity of TIE based on SD3.5-Large.** We report BLIP-VQA scores under different choices of γ, λ_{ext}, λ_{int}, and $\mathcal{C}$. The default configuration is highlighted in gray. Best results are shown in **bold** and second-best results are underlined.

Hyperparameters				BLIP-VQA ↑					
γ	λ_{ext}	λ_{int}	$\mathcal{C}$	Color	Shape	Texture	Spatial	Complex	Avg.
0.001	0.1	0.1	0.5	0.7804	0.6174	0.7577	0.6931	0.5128	0.6723
0.001	0.2	0.1	0.5	0.7806	0.6174	0.7573	0.6927	0.5124	0.6721
0.001	0.3	0.1	0.5	0.7806	0.6167	0.7574	0.6925	0.5124	0.6719
0.001	0.2	0.2	0.5	0.7812	**0.6196**	0.7572	0.6927	0.5118	0.6725
0.001	0.2	0.3	0.5	0.7823	0.6173	0.7560	0.6926	0.5105	0.6717
0.001	0.2	0.4	0.5	**0.7847**	0.6168	0.7548	0.6927	0.5113	0.6721
0.001	0.2	0.1	0.4	0.7792	0.6167	0.7561	0.6921	0.5129	0.6714
0.001	0.2	0.1	0.6	0.7810	0.6163	**0.7591**	0.6923	0.5121	0.6722
0.0005	0.1	0.1	0.5	0.7795	0.6192	0.7544	**0.6933**	0.5165	0.6726
0.0005	0.1	0.2	0.5	0.7826	0.6185	0.7548	**0.6933**	0.5154	**0.6729**

ing, and that the degree of binding should be adaptively adjusted based on each concept's attention strength.

Comparison of SDXL and TIE Using Traditional Similarity Metrics. In Sec. 3.2, we reveal that CLIP text embeddings indicate only a subtle distinction between related and unrelated concepts using traditional similarity metrics. To further examine this limitation, we analyze the distributions of cosine similarity and L2 distance across additional datasets and assess how they change when applying our TIE. As shown in Fig. 3, TIE text embeddings more clearly differentiate between related and unrelated concepts compared to the original SDXL embeddings across both datasets.

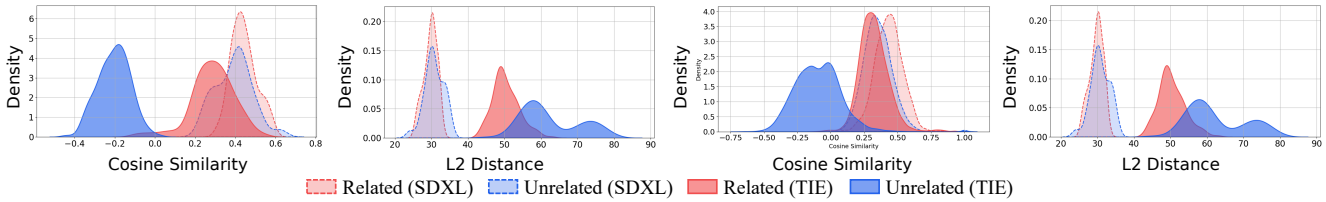
E.2. Entropy Predictive Validity

We provide additional analyses of how token entropy relates to concept rendering failures. Unless otherwise specified, the following results are obtained with CLIP-G.

Entropy across rendering outcomes. We first examine whether entropy systematically increases as compositional rendering deteriorates. For each generated image, we average the entropy of its concept tokens and group the samples according to whether both concepts are rendered, only one is rendered, or neither is rendered. The mean entropy increases from 0.9496 when both concepts are rendered ($n = 1{,}696$), to 0.9511 when one concept is missing ($n = 245$), and to 0.9518 when neither concept is rendered ($n = 27$). The difference between the fully rendered and fully missing groups is $+0.0022$, with a 95% bootstrap confidence interval of $[+0.0012, +0.0033]$ using 2,000 prompt-level bootstrap replicates ($d = 1.12$; Kruskal–Wallis $p = 1.6 \times 10^{-30}$). Thus, higher entropy is consistently associated with more severe compositional failures.

Predictive validity and calibration. At the prompt level, mean concept entropy discriminates prompts with neither concept rendered from those with both concepts rendered

Figure 2. **L0 attention scores and generated images.** The underline indicates the concept corresponding to each L0 attention score range.



(a) A&E Data Distribution (b) T2I-Compbench Data Distribution

Figure 3. **Comparison of traditional similarity metrics between SDXL [12] and TIE**

with an AUC of 0.810 (95% CI $[0.691, 0.895]$). For the broader distinction between any concept loss and fully successful rendering, the AUC is 0.718. At the finer level of predicting which individual concept will fail, the AUC is lower at 0.569 (95% CI $[0.549, 0.591]$), and remains $0.599/0.561$ when evaluated separately within the two concept-position strata. This distinction is expected: prompt-level entropy captures whether a prompt is prone to compositional failure, but is not intended to serve as a binary classifier for identifying the exact concept that will be lost.

The same trend is observed when concepts are grouped by entropy quintile. The concept-loss rate increases from 4.5% in the lowest-entropy quintile to 34.6% in the highest-entropy quintile, with a strong positive rank trend ($r = 0.81$). The relationship is not strictly monotonic across every adjacent bin (the third quintile decreases to 5.9%), but the overall increase across entropy levels is pronounced. Moreover, a prompt-clustered logistic regression controlling for token position shows that a one-standard-deviation increase in entropy is associated with $1.30\times$ higher odds of concept loss ($p = 1.1 \times 10^{-10}$). Adding entropy significantly improves over a position-only model ($\chi^2 = 33.7$). Together, these results support our use of token entropy as a *continuous proxy for concept-loss propensity*, rather than as a deterministic failure classifier.

Causal intervention along the shared CLIP direction. We further test whether the geometric factor associated with entropy has a causal effect on concept loss. As described in the main paper, token entropy is strongly correlated with alignment to CLIP's shared anisotropic direction $\hat{\mu}$. We therefore apply an exact norm-preserving rotation in the plane spanned by a concept embedding c_i and

$\hat{\mu}$, while keeping the displacement magnitude fixed. Starting from a concept-loss rate of 27.3%, rotating the embedding toward $\hat{\mu}$ increases the loss rate to 35.7% (McNemar $p = 1 \times 10^{-4}$), whereas rotating it away from $\hat{\mu}$ reduces the loss rate to 22.3% ($p = 0.007$). The intervention preserves the embedding norm up to numerical precision ($\max |\Delta\|c_i\|_2| = 7.6 \times 10^{-6}$). In contrast, displacement-matched rotations toward orthogonal or random-coordinate directions change the loss rate by at most 0.3 percentage points ($p = 1.00$). These controlled interventions show that concept loss is specifically sensitive to movement along CLIP's shared direction, rather than to embedding perturbation magnitude alone.

Replication on SDXL. We additionally repeat the predictive-validity analysis on SDXL using 492 prompts and 1,968 generated images. The entropy difference between successful and failed rendering remains significant ($+0.0007$, $p = 4.2 \times 10^{-8}$), with a prompt-level AUC of 0.603. The calibration trend is also strictly monotonic across all five entropy quintiles, with the concept-loss rate increasing from 38.4% to 53.0%. Although the effect is smaller than for SD3.5, this is consistent with the substantially higher base concept-loss rate of SDXL.

E.3. Scope Beyond CLIP Text Encoders

We further examine whether the positional entropy pattern observed in CLIP generalizes to non-CLIP text encoders. Specifically, we repeat the position-controlled analysis using Qwen2.5-VL, Gemma-2-2B, Llama-3-8B, and T5. Although these encoders also exhibit anisotropic embedding geometry, the later-position increase in token entropy does not consistently transfer beyond CLIP. Using only nouns observed in both prompt positions, CLIP-G shows a significant entropy increase of $+0.0033$ at the later position ($d_z = 1.08$, $p = 1.9 \times 10^{-9}$), whereas none of the tested non-CLIP encoders exhibits a significant corresponding effect after multiple-comparison correction. These results motivate restricting our analysis and TIE to CLIP-conditioned T2I models rather than claiming encoder-independent generalization.

F. Additional Ablation Studies

Effect of each component of TIE. We further ablate each component of TIE based on SD3.5-Large by setting its corresponding coefficient to zero, and report BLIP-VQA in Tab. 3. Removing any single component degrades the average score, while the full configuration attains the best overall performance. Although disabling a component occasionally improves an individual attribute—e.g., $\gamma = 0$ yields the highest Color and Texture scores—it consistently comes at the cost of the remaining attributes, indicating that the components are complementary rather than redundant.

Table 3. **Effect of each component of TIE.** Each row disables one component by setting its coefficient to zero. Text., Spat., and Comp. denote texture, spatial, and complex, respectively. Best results are shown in **bold** and second-best results are underlined.

	BLIP-VQA [8] ↑					
Method	Color	Shape	Text.	Spat.	Comp.	Avg.
w/o γ	**0.7831**	0.6137	**0.7550**	0.6923	**0.5171**	0.6722
w/o λ_{ext}	0.7816	0.6173	0.7546	0.6931	0.5148	0.6723
w/o λ_{int}	0.7773	0.6166	0.7531	**0.6933**	0.5166	0.6714
TIE (Ours)	0.7826	**0.6185**	0.7548	**0.6933**	0.5154	**0.6729**

Table 4. **The effect of operation sequence of TIE.** Config.1 refers to Sequential IEB → AEP and Config.2 is Sequential Extrapolation → Interpolation. We measure BLIP-VQA [8] for when the order of AEP and IEB are switched and when the order of interpolation and extrapolation is switched. Best results are shown in **bold** and second-best results are underlined.

Method	BLIP-VQA [8] ↑			
	Color	Shape	Texture	Avg.
Config.1	0.6995	**0.5636**	0.6700	0.6444
Config.2	0.6253	0.5329	0.6232	0.5938
TIE (Ours)	**0.7004**	0.5632	**0.6703**	**0.6447**

AEP without Entropy-aware Boost. In Fig. 4, some qualitative examples show the difference between our method without taking the token entropy into account. This often causes an over-manipulation of the tokens as the model is not aware of whether the concepts are preserved or not. Thus, it is important for the model to be entropy-aware, so that it preserves concepts effectively.

IEB without L0 Attention Scores. As shown in Fig. 5, applying IEB without incorporating L0 attention scores leads to two major issues. First, it can result in overmanipulation. When constant scaling is applied, the generated image may deviate from the intended semantics, causing artifacts such as additional objects or an unnatural painting-like appearance. Next, concept confusion persists where objects are merged or attribute bindings remain incorrect. These observations underscore the importance of incorporating L0 attention scores into IEB to enable more precise manipulation.

G. Additional Experiments

Additional Quantitative Results. We further evaluate TIE using complementary metrics beyond BLIP-VQA to assess both compositional faithfulness and perceptual quality. We recruit 15 participants who view 9 prompts (3 each

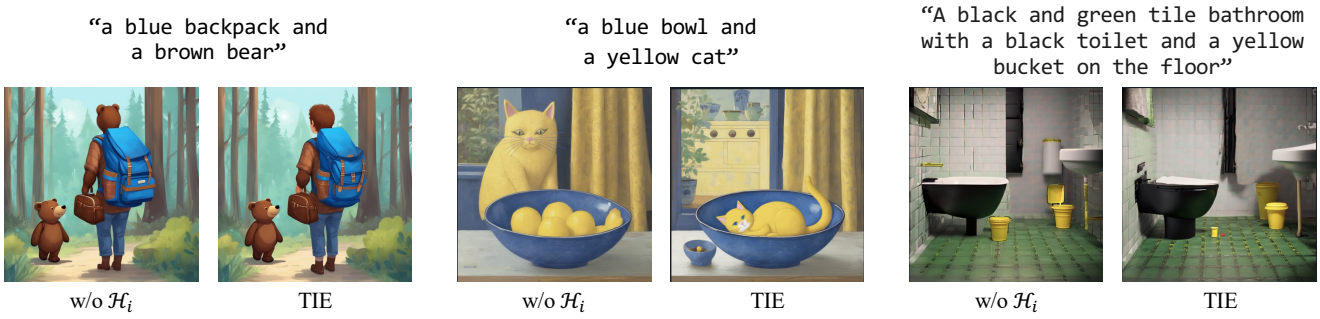


Figure 4. **Comparison between our method without entropy and TIE.**

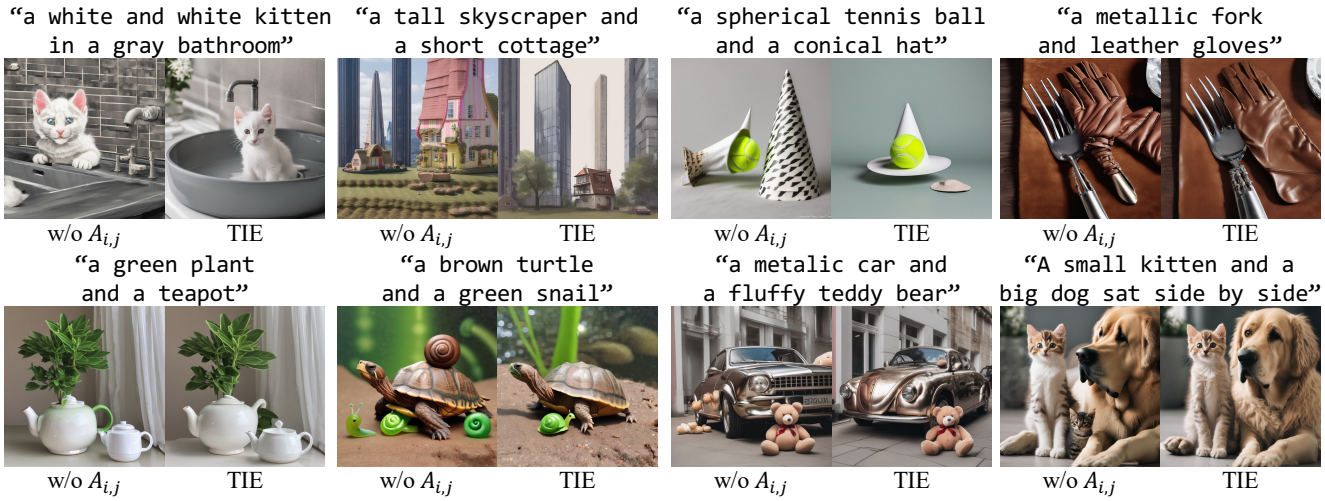


Figure 5. **Comparison between our method without L0 attention scores and TIE.**

from the color, shape, and texture categories) and, for each prompt, rate the outputs of all 6 compared methods from 1 (worst) to 5 (best) on *perceptual quality* and *semantic alignment* with the prompt. Within each prompt, images are presented in randomized order and method identities are withheld from participants. As shown in Tab. 6, TIE obtains the highest mean rating on both criteria, with the larger margin on semantic alignment, indicating that strengthening compositional binding does not degrade visual quality. We also evaluate object-level correctness on CC-500 using NoctOwl [1], where TIE attains the highest detection score of 0.6146 while maintaining comparable detection confidence, showing that the gains also hold under a detector-based protocol rather than a VQA-based one.

Statistical Robustness across Random Seeds. We further evaluate the robustness of TIE across five matched random seeds on SD3.5-Medium. For each seed, we compute the overall BLIP-VQA score averaged across the five T2I-CompBench categories using the same prompts for the baseline and TIE. As shown in Tab. 5, TIE improves the overall score from 0.6562 ± 0.0072 to 0.6652 ± 0.0049. The paired gain is $+0.0090$ with a 95% confidence interval of $[+0.0056, +0.0124]$. A paired t-test confirms that the improvement is statistically significant ($t(4) = 7.361$, $p = 0.0018$), indicating that the gain is consistent across random seeds.

Table 5. **Statistical robustness across five matched random seeds.** We report the overall BLIP-VQA score averaged across the five T2I-CompBench categories on SD3.5-Medium.

Method	BLIP-VQA ↑	Paired Gain
SD3.5-Medium	0.6562 ± 0.0072	–
TIE (Ours)	$\mathbf{0.6652\pm0.0049}$	$\mathbf{+0.0090}$

Additional Evaluation on GenEval2. We further evaluate TIE on GenEval2 using a diverse set of VQA-, alignment-, and reward-based metrics. As shown in Tables 7 and 8, TIE improves over the SD3.5-Medium backbone on VQAScore, TIFA, both Soft-TIFA variants,

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Table 6. Evaluation on perceptual quality, semantic alignment, and CC-500. We compare TIE with existing compositional generation methods in terms of perceptual quality, semantic alignment, and detection and confidence scores on CC-500 using NoctOwl [1]. Best results are shown in **bold** and second-best results are underlined.

Method	Perceptual Quality ↑	Semantic Alignment ↑	CC-500 (NoctOwl) ↑	
			Det.	Conf.
SDXL	3.80	2.82	0.4134	0.4287
A&E$_{XL}$	3.81	2.81	0.5017	**0.4661**
Syngen$_{XL}$	3.39	2.81	0.3970	0.4196
ToMe	3.28	2.46	0.3519	0.4135
Magnet	3.64	2.89	0.5208	0.4338
TIE (Ours)	**3.88**	**4.56**	**0.6146**	0.4307

Table 7. **Quantitative comparison on GenEval2.** We compare TIE with recent inference-time methods on SD3.5-Medium. Best results are shown in **bold** and second-best results are underlined.

Method	VQAScore ↑	TIFA ↑	Soft-TIFA$_{AM}$ ↑	Soft-TIFA$_{GM}$ ↑
SD3.5-M	19.73	65.73	65.99	19.94
TACA	20.90	66.84	66.00	20.73
FOCUS	**21.50**	**67.60**	**67.78**	**20.96**
Magnet	20.68	66.05	66.23	20.77
TIE (Ours)	20.21	66.17	67.00	20.10

SigLIP-2, BLIP, Qwen, and ImageReward, while showing slightly lower CLIP and PickScore scores. Among the compared methods, TIE achieves the second-best Soft-TIFA$_{AM}$ and BLIP scores and the best Qwen-based alignment score, although FOCUS and TACA perform better on several VQA-based metrics. Thus, while TIE does not uniformly outperform these methods, which optimize at every denoising step at 1.4–4.3× the inference cost, it provides a competitive single-update, sampling-free alternative.

The relative ranking across methods also varies with the evaluator. In particular, TIE shows strong improvements under the BLIP-VQA evaluation used in our main experiments, whereas its advantage is less pronounced under alternative VQA and reward models. Nevertheless, TIE remains broadly competitive across these complementary metrics and consistently improves over the backbone on most evaluators, without systematic degradation in general text–image alignment. These results suggest that the benefits of TIE are robust but evaluator-dependent rather than uniformly dominant across all metrics.

Reverse the Order of AEP and IEB. We look into the order of the methods in TIE and present our results in Table 4. In TIE, AEP is applied before IEB and interpolation is always applied ahead of extrapolation. We switch the order of these methods to show the effectiveness of our method TIE. This experiment was conducted with SDXL [13] backbone.

Table 8. **Quantitative comparison on GenEval2 using alignment and reward metrics.** We compare TIE with recent inference-time methods on SD3.5-Medium. Best results are shown in **bold** and second-best results are underlined.

Method	CLIP I-T ↑	SigLIP-2 I-T ↑	BLIP T-T ↑	Qwen T-T ↑	PickScore I-T ↑	ImgRew I-T ↑
SD3.5-M	0.3551	0.2523	0.7413	0.5766	23.025	1.4068
TACA	0.3553	0.2554	**0.7424**	0.5685	**23.0915**	1.4509
FOCUS	**0.3554**	**0.2560**	0.7410	0.5715	22.9023	**1.4617**
Magnet	0.3551	0.2527	0.7411	0.5748	23.018	1.4163
TIE (Ours)	0.3548	0.2528	0.7418	**0.5775**	23.011	1.4101

Swapping AEP and IEB (Config.1) yields a minor decrease in Color and Texture scores and a slight uptick in Shape. This demonstrates that beginning with AEP to strengthen each token to prevent concept loss *before* binding the related tokens to alleviate concept confusion is effective.

Reverse the Order of Interpolation and Extrapolation. When extrapolation is applied before interpolation (Config.2), all three metrics drop sharply. This shows that applying interpolation first is essential to establish a stable attribute-specific embedding distribution before repelling the tokens that are unrelated to that specific token.

H. Limitations

TIE is currently limited to CLIP-conditioned T2I models, as the positional entropy effect does not consistently transfer to the tested non-CLIP text encoders. It also cannot fully overcome inherent limitations of T2I models, such as biases inherited from training data. Extending the proposed signals to other text encoders remains an important direction for future work.

I. Additional Qualitative Results

We provide additional results on the CC-500 dataset in Fig. 7 and Fig. 8. We also provide additional results on the T2I-Compbench dataset in Figs. 9–10 for SDXL and Figs. 11–15 for SD3.5.

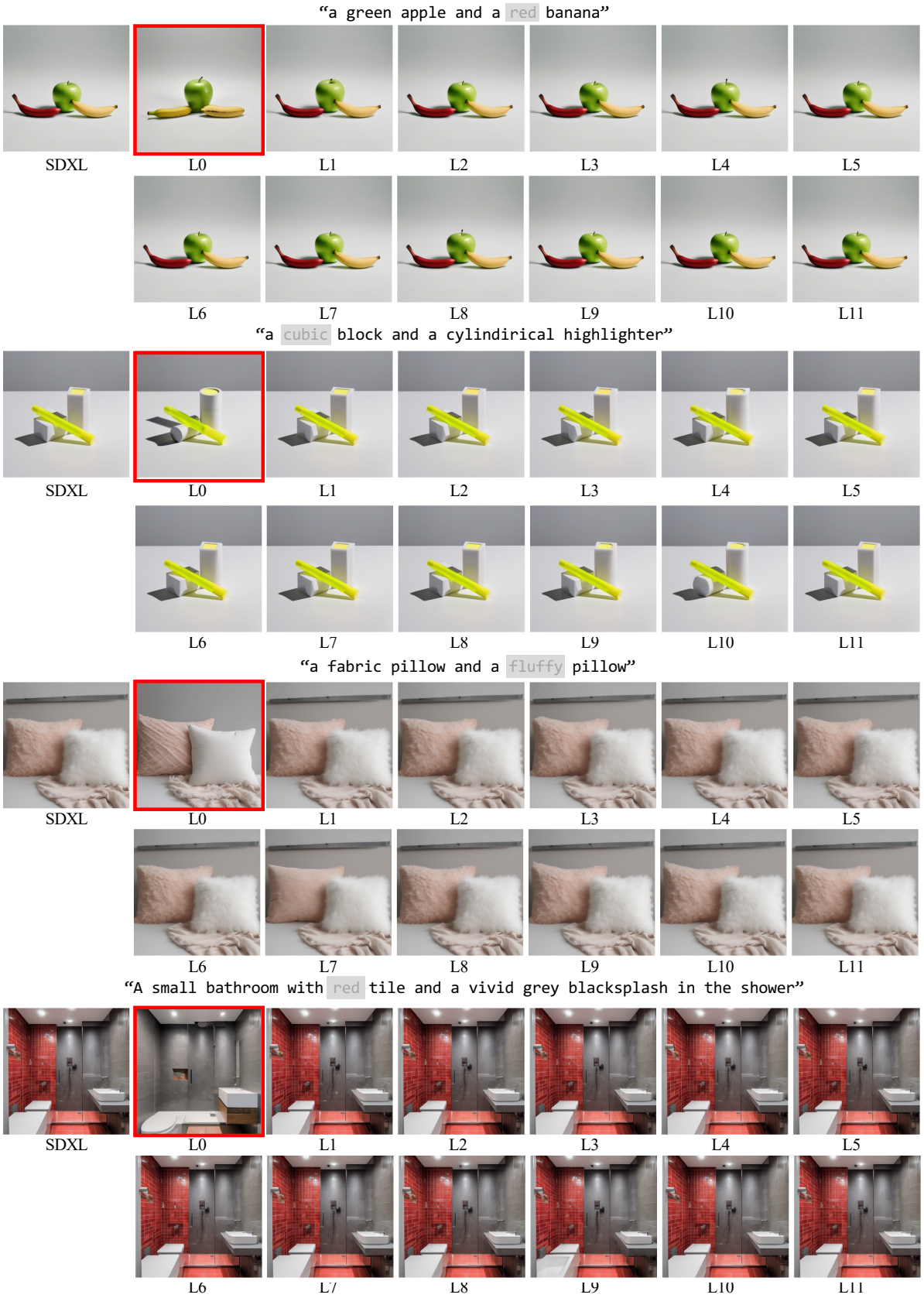

Figure 6. **Results of masking L0 attention scores in SDXL.** The gray box in the prompt indicates the masked attribute, and the red outline shows the result after applying masking at Layer 0.

Figure 7. **Qualitative results based on SDXL [12] from the CC-500 dataset.**

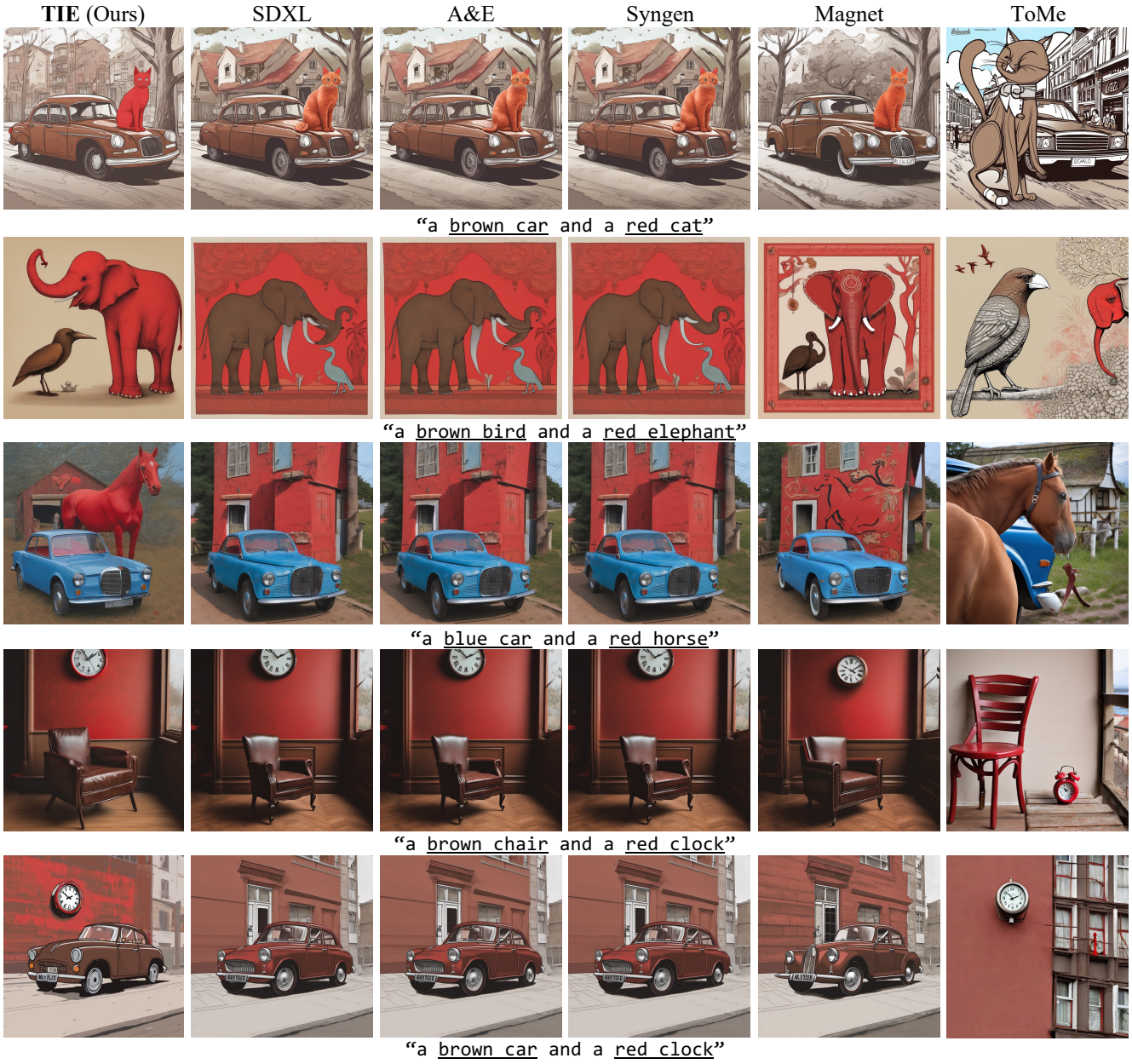

Figure 8. **Qualitative results based on SDXL [12] from the CC-500 dataset.**

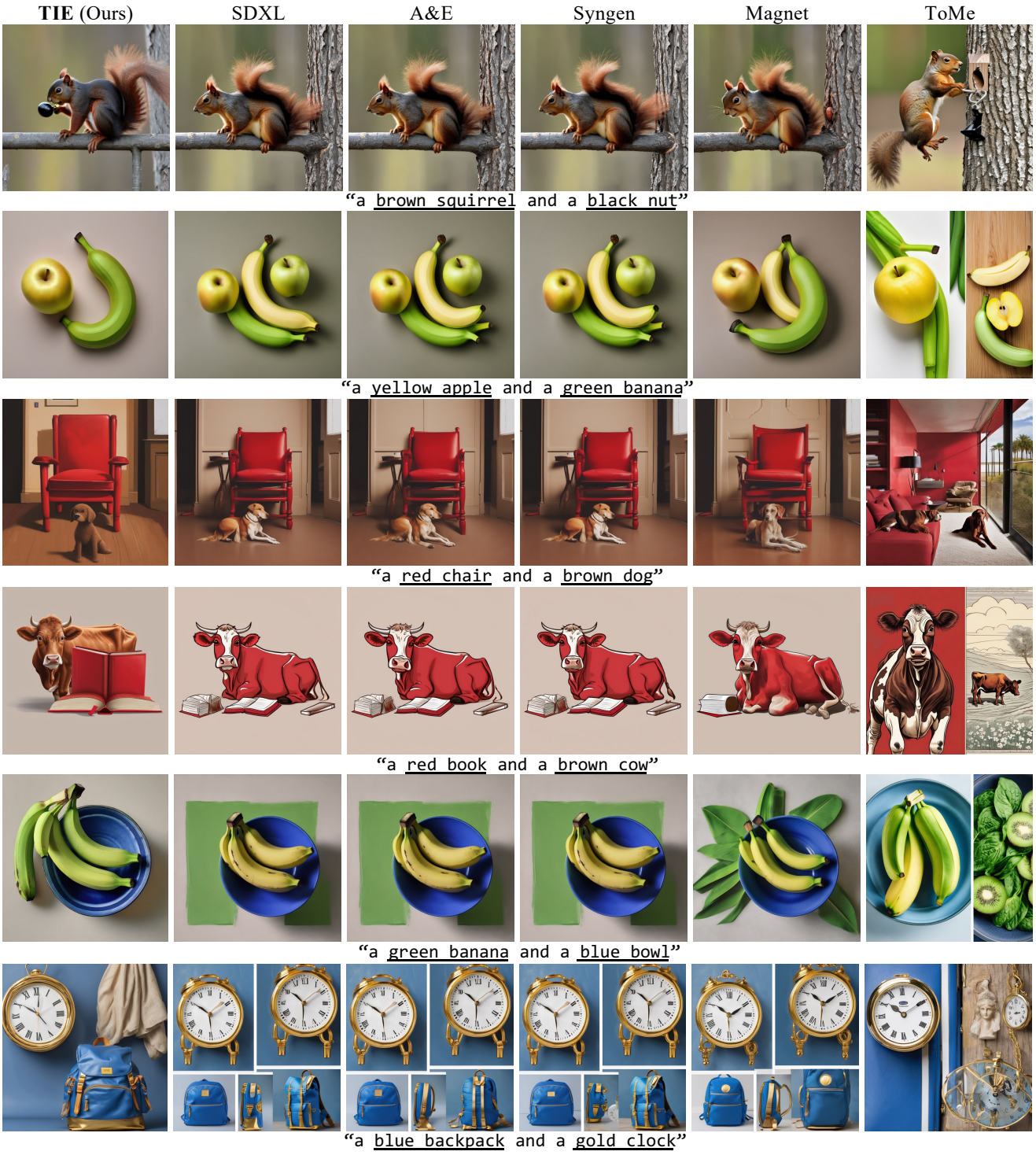

Figure 9. **Qualitative results based on SDXL [12] from the T2I-Compbench dataset Color prompts.**

Figure 10. **Qualitative results based on SD3.5 [3] from the T2I Compbench dataset.**

Figure 11. **Qualitative results based on SD3.5 [3] from the T2I Compbench dataset Color prompts.**

Figure 12. **Qualitative results based on SD3.5 [3] from the T2I Compbench dataset Shape prompts.**

Figure 13. **Qualitative results based on SD3.5 [3] from the T2I Compbench dataset Texture prompts.**

Figure 14. **Qualitative results based on SD3.5 [3] from the T2I Compbench dataset Spatial prompts.**

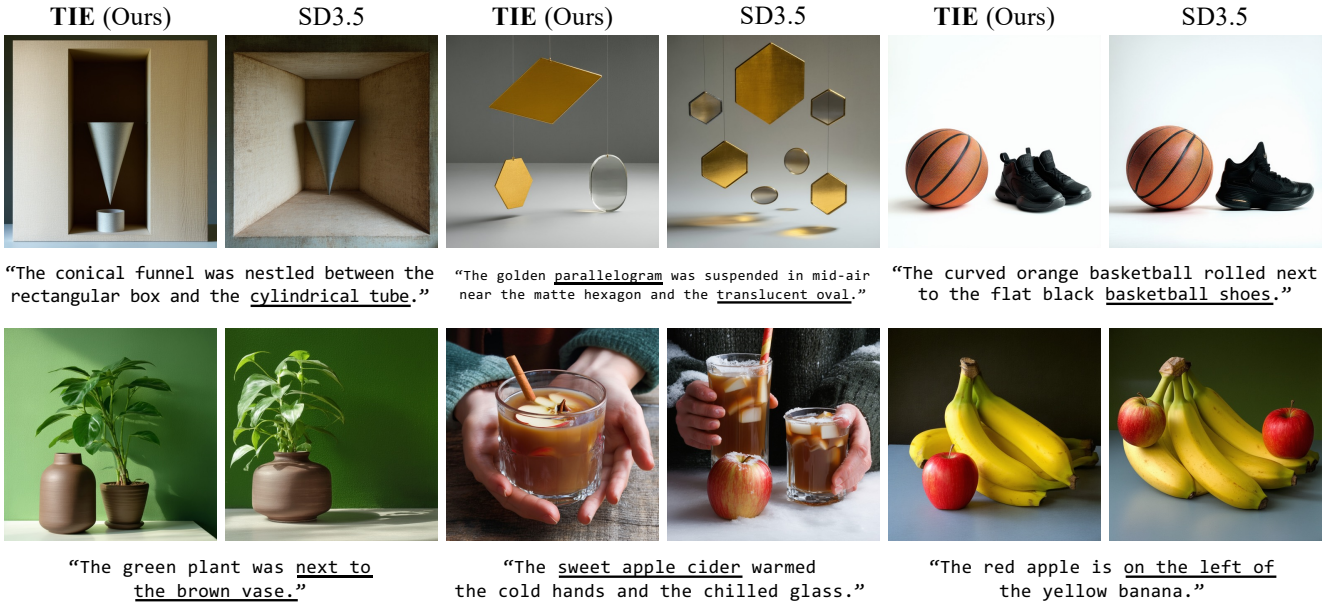

Figure 15. **Qualitative results based on SD3.5 [3] from the T2I Compbench dataset Complex prompts.**

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